INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

SAWS, BAND, CUTOFF, HORIZONTAL, NON-ADJUSTABLE AND ADJUSTABLE CUTTING ANGLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heavy duty, horizontal type, production metal cutting band saw machines equipped for manual stock indexing (type I) or for automatic, repetitive indexing of bar material to preset lengths (type II). Class A machines are capable of square, 90 degree cutoff. Class B machines are capable of miter sawing from 0 degree to 45 degrees cutting angle. The machines shall be suitable for cutting-off a variety of metal bars including, but not limited to, solid bars, tubing, pipe, structural shapes, and extrusions within the capacity of the machine. The requirements set forth in this specification and the completed ordering data (see paragraph 6.2), establishes the Governments minimum requirements for each procurement.

1.2 Classification. The band saw shall be of the following types, classes, and sizes. The type, class, and size to be furnished shall be as specified (see 3.5 and 6.2).

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3405

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Type I - Non-automatic - Manual feed

Class A - Non-adjustable cutting angle
(90 degrees from workpiece axis)

Size 1 - 12 by 12-inch workpiece capacity
Size 2 - 12 by 16-inch workpiece capacity
Size 3 - 12 by 20-inch workpiece capacity
Size 4 - 14 by 17-inch workpiece capacity
Size 5 - 15 by 20-inch workpiece capacity

Class B - Adjustable cutting angle (90 degrees to 45 degrees
from workpiece axis)

Size 2 - 12 by 16-inch workpiece capacity

Type II - Automatic - Powered stock feed

Class A - Non-adjustable cutting angle
(90 degrees from workpiece axis)

Size 1 - 12 by 12-inch workpiece capacity
Size 2 - 12 by 16-inch workpiece capacity
Size 3 - 12 by 20-inch workpiece capacity
Size 4 - 14 by 17-inch workpiece capacity
Size 5 - 15 by 20-inch workpiece capacity

Class B - Adjustable cutting angle (90 degrees to 45 degrees
from workpiece axis)

Size 2 - 12 by 16-inch workpiece capacity
Size 4 - 14 by 17-inch workpiece capacity

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S.DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS

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cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.10 - Safety Requirements for the Construction, Care and Use of Metal Sawing Machines. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Design. The band saw shall be new (not a prototype) and one of the manufacturers current production models capable of operations in accordance with the requirements herein. The band saw shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.4.1), metric (see 6.4.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the band saw shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than .01 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of use, rebuilt or remanufactured products are allowed under this specification.

3.2.3 Energy efficiency. The band saw and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the band saw, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.10. If a conflict arises between 29 CFR 1910 and ANSI B11.10, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial band saws supplied to the commercial market shall be

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provided for the point of operation and all other parts of the band saw that present safety hazards. Additional guarding dependent on workpiece size and configuration shall be provided when specified and fully described by the procuring activity (see 6.2).

3.2.6 Mercury restriction. The band saw shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the band saw.

3.2.8 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable and shall have a low lubrication level warning light that is in clear view of the operators normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. To provide for replacement or worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The band saw shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the band saw to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only when specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.3.5 Painting. Unless otherwise specified (see 6.2), the band saw shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the band saw and its related attachments and accessories shall conform to FED-STD-H28 and the applicable " Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 2000.A88 and 390.03 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The band saw shall consist of a base, frame, band drive system, band wheels, band guides and guide arms, vise, hydraulic system, coolant system and such other components as are necessary to accomplish the performance requirements described herein.

3.4.1 Base.

3.4.1.1 Type I or II, class A. The base of type I or type II, class A machines shall be of heavy duty construction and shall support the vise, chip collector, coolant reservoir, coolant pump, hydraulic system, sawing frame and workpiece. Hinged access doors or removable panels shall be provided to facilitate servicing and adjusting internal components. Means shall be provided for leveling and securely anchoring the machine.

3.4.1.2 Type I or II, class B. The base of the type I or II, class B machines shall conform to all requirements for class A machines. In addition, either the sawing frame or the base upon which the saw frame is mounted shall swivel from 90 degrees to 45 degrees in relation to the work axis. Swiveling shall be accomplished by power. A graduated scale shall be provided for setting the angle of cut. Means shall be provided to securely lock the angle of cut. Means shall be provided to securely lock the swivel unit in set position for sawing.

3.4.2 Sawing frame. The frame unit shall contain the band guides and guide arms, automatic band tension device and saw band. A protective cover shall enclose the band wheels and saw blade so that a minimum of blade length is exposed for sawing. The cover, or guard, shall have hinged doors or other suitable means for providing ready access to the blade wheels. A hydraulic system shall furnish power for raising and lowering the sawing frame and maintaining constant feed force. A pressure gauge or other device shall be provided to indicate the down feed pressure of the sawing frame.

3.4.3 Band drive system. The drive system shall provide infinitely variable band speeds. Gears or gear shafts utilized shall be supported by ball or roller

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bearings in completely enclosed housings, and shall have positive means of lubrication. A band speed indicator shall be provided to indicate speed of the band in feet per minute.

3.4.4 Band feed system. The machine shall have a band feed system capable of providing infinitely variable feed forces. Operators control devices shall be provided for selecting down feed, neutral, and retract functions over a range of 0 to not less than 300 pounds.

3.4.5 Band wheels. The band wheels shall be the flanged type. The idler wheel shall have a metal face or replaceable resilient tire and shall be adjustable to facilitate changing saw bands and to accommodate varying saw band lengths. The band wheels shall be protected from chips by a suitable band cleaning brush which is installed to clean chips from the saw band. The wheels shall be supported by ball or roller bearings which are either continuously lubricated or the sealed-for-life-type.

3.4.6 Band tensional devices. An adjustable tensioning mechanism shall be installed on the idler wheel, capable of maintaining a band tension of up to 30,000 psi either mechanically or hydraulically while the machine is in operation. Band tension may be applied automatically, or may be manually adjusted by turning a hand wheel to a marked setting. Automatically applied band tensioning mechanisms shall maintain a preset force while cutting, and relax band tension when stopped, without dislocating the blade on the wheels.

3.4.7 Band guides and guide arms. Band guides shall consist of solid carbide or carbide faced inserts, hardened and ground side rollers on each side of the saw band, or a combination of these, and a carbide back-up block on roller mounted or ball bearings. Two guide arms shall be attached to the sawing head for supporting the band guides which, in turn, provide the rigidity and stability required for the machine to perform production cutoff operations. One or both of the arms shall be adjustable for cutting various workpiece widths within the range of the machine. The adjustable arm(s) shall be easily positioned and shall be provided with locking mechanisms.

3.4.8 Worktable and vise. The worktable and vise shall be capable of supporting workpiece(s) within the rated capacity of the machine. The vise shall be power operated and shall have provision for adjusting the clamping pressure on the workpiece.

3.4.9 Automatic bar feed unit. Type II saws shall have an automatic stock feeding device, conforming to this paragraph, fitted to the saw. The automatic feeding device shall have a direct reading digital length readout calibrated in either inches or millimeters, or in combination. The automatic feeding device shall include a manual feature to make a single cut and a feature to accommodate multiple cuts over an incremental range of not less than 0-12 inches per single index feature to produce workpieces longer than 12 inches and up to 72 inches in length. The ways of feeding devices shall be either covered, enclosed, or have wipers and be lubricated by either a manual or automatic lubricating system. Automatic feeding devices shall include the number of power operated vises necessary to feed stock material (forward and reverse) through the automatic cycles to preclude manual push-up of the stock. When specified (see 6.2), the bar feed unit shall be equipped with a cycle counter which allows the operator to preset at the number of pieces to be cut before automatic machine shut down.

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3.4.10 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.11 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.12 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hertz (Hz) circuit. Unless otherwise specified (see 6.2), the band saw shall be initially wired for operation on 460 volts.

3.4.12.1 Motors. Motors shall meet the requirements of 3.4.12. Motors shall be rated for 60-minute duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (ac) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.12.2 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.12.3 Hour meter. When specified (see 6.2,) the band saw shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time for the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the band saw.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the band saw shall meet the requirements of table I for the machine specified.

3.6 Performance.

3.6.1 Metal removal rate and accuracy. The machine shall be capable of sawing SAE 1020 steel at a rate of 15 square inches per minute when tested in accordance with 4.7.4. Saw cuts produced at these rates shall be perpendicular to the surface resting on the worktable within +/- 0.002 inch per inch of thickness and shall be flat within +/- 0.002 inch per inch (concave or convex).

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3.6.2 Automatic bar feed accuracy. Type II machines shall be capable of cutting the disks in 4.7.4.2 within +/- 0.005 inch of preset length. The thickness of the 12 disks shall vary not more than 0.010 inch.

3.6.3 Angular cut accuracy, class B machines. Class B machines shall be capable of cutting angles to within one-half degree of the selected angle to which the machine is set.

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.9 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the chart or plate shall be as listed below.

Points of lubricate application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below.

Nomenclature
Manufacturers name
Manufacturers model designation
Manufacturers serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE or Military or Federal specification number)
Type of filter

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3.11 Workmanship. Workmanship of the band saw and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturer's producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the first article examination and all test to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The band saw shall be examined to determine compliance with all requirements of this specification.

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4.7 Test.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104 degrees Fahrenheit and 20 to 95 percent relative humidity.

4.7.2 Operational test. The band saw shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.7.4 through 4.7 4.1. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.7.4 Performance tests. The machine shall be subjected to the following tests to determine compliance with the performance and accuracy requirements of 3.6 for the purpose of these tests. Sawing time shall include time consumed during stalls, but shall exclude time required for band changes. A high speed bi-metal saw band shall be used for all tests.

4.7.4.1 Bar stock sawing test. Each machine shall make 12 cuts through 5-inch diameter, SAE 1020 steel bar stock. The disk produced by the cuts shall be not less than 3/8-inch thick. The cutting speed rate and feed rate shall be at the option of the manufacturer. The time consumed while making the twelve cuts shall be determined and the sawing rate calculated using the actual cross section of the bar. The sawing rate, flatness, and squareness of the cut shall conform to 3.6.1.

4.7.4.2 Automatic bar feed accuracy test. Each machine provided with an automatic bar feed unit shall perform the test in 4.7.4.1 utilizing the automatic bar feed to make twelve cuts. Each disk cut shall meet the requirements of 3.6.2. On machines equipped with a cut counter, the counter shall be set for 12 cuts and the machine shall automatically stop after 12 disks have been cut.

4.7.4.3 Angular cut accuracy test (class B machines). Using the same type workpiece described in 4.7.4.1, the machine base or sawing frame, shall be swiveled and a cut shall be made at 15, 30, and 45 degrees. Each angular cut shall meet the requirements of 3.6.3.

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

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6. NOTES

6.1 Intended use. Machines covered by this specification are intended for use in heavy duty, production, high speed cutoff operations. Class A machines provide straight cutoff at 90 degrees to workpiece axis. Class B machines provide for straight (90 degree) cutoff or angular (0 to 45 degree) cutoff to the workpiece axis.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Bar feed cycle counter for type II machines if required (see 3.4.9).

k. Electrical system, if different (see 3.4.12).

l. Initial voltage, if different (see 3.4.12).

m. Motors, if different (see 3.4.12.1).

n. Worklight, if not required (see 3.4.12.2).

o. Hour meter, if required (see 3.4.12.3).

p. Size and capacity, if different (see 3.5).

q. Specify and fully describe optional equipment, if required (see 3.8).

r. Specify applicable repair parts, if required (see 3.9).

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s. Lubrication plate, if required (see 3.10.1).

t. Information to be provided on lubrication plate, if different (see 3.10.1).

u. Nameplate, if different (see 3.10.2).

v. Hydraulic plate, if required (see 3.10.3).

w. Inspection conditions, if different (see 4.3).

x. First article inspection, if different (see 4.4).

y. Quality conformance inspection, if different (see 4.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data , except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Formal 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.4.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International D'Units (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-376).

6.5 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

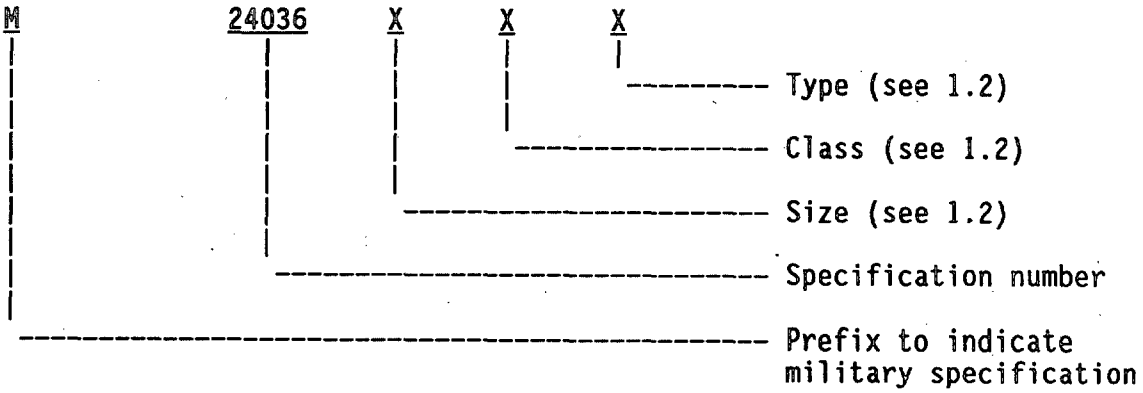
6.8 Training. Training required by the procuring activity should be provided as specified in the contract.

6.9 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.10 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

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6.11 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.12 Subject term (keyword) listing.

90 degree cutting
0 to 45 degree cutting
Metal pipe
Solid metal bars
Tubing

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TABLE I
Sizes and capacities.

Manual Stock Feed

Type I - Class A and Class B

Characteristics	Size 1	Size 2	Size 3	Size 4	Size 5
Workpiece capacity, in:					
Rectangular	12 x 12	12 x 16	12 x 20	14 x 17	15 x 20
Round	12	12	12	14	15
Band data:					
Width & thickness:	1 1/4 X .042	1 1/4 X .042	1 1/4 X .042	1 1/4 X .042	1 1/4 X .042
Band drive motor, hp:	5	5	5	5	5
Optional hp:	-	-	10	10	10
Band feed force, infinitely variable:					
Pounds:	0 - 300	0 - 400	0 - 600	0 - 300	0 - 400
Optional:	-	-	-	-	0 - 600
Band speed range, infinitely variable, RPM:	60 - 300	60 - 350	60 - 350	60 - 450	55 - 400

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TABLE I, cont'

Type II - Class A, and Class B

Automatic - Powered Stock Feed

Characteristic	Size 1	Size 2	Size 3	Size 4	Size 5
Workpiece capacity, in:					
Rectangular:	12 X 12	12 X 16	12 X 20	14 X 17	15 X 20
Round	12	12	12	14	15
Band data:	1-1/4	1-1/4	1-1/2	1-1/4	1-1/4
Width & thickness:	X	X	X	X	X
	.042	.042	.050	.042	.042
Band drive motor, hp:	5	5	5	5	10
Band feed force, infinitely variable:					
Pounds:	0 - 300	0 - 400	0 - 600	0 - 300	0 - 400
Optional:	—	—	—	—	0 - 600
Band speed range infinitely variable:					
RPM:	0 - 350	60 - 350	60 - 350	60 - 450	60 -400

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered, nor greater than the requirements of the next larger size shown in above table. When a larger size and capacity machine is offered, the physical dimensions of the machine shall not exceed that of the stated machine by mote than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84

User activities:
Army - SM
Navy - MC

Preparing activity:
DLA-IP

(Project Number 3405-0071)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-24036E	Saws, Band, Cutoff, Horizontal, Non-Adjustable and Adjustable Cutting Angle

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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